David Chang

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Education

University of Texas at Austin **Aug 2023 – Expected May 2028**
Bachelor of Science in Computer Science Austin, TX
Bachelor of Science in Electrical and Computer Engineering GPA: 3.6/4.0

Skills

Programming Languages: ARM Assembly, C, C++, C#, Java, JavaScript, Python, TypeScript

Relevant Tools: AWS, Docker, Kubernetes, Linux, FastAPI, PostgreSQL, SQLite, RDF/SPARQL, Git, .NET, Django, React

Coursework: Computer Architecture, Data Structures and Algorithms, Discrete Math, Embedded Systems, Linear Algebra

Experience

Ajou University School of Medicine **Jun 2024 – Sep 2024**
Intern | Clinical AI Research *Suwon, South Korea*

- Co-authored **2 peer-reviewed journal papers** (**11 citations**) by building BERT/GPT-based clinical NLP pipelines.
- Reduced OHDSI's Common Data Model ETL runtime by **23%** by migrating R pipelines to Python and replacing repeated row-wise operations with hash-join-based transformations.

LooxidLabs **Jun 2022 – Jan 2023**
Intern | Product Development & Analytics *Seoul, South Korea*

- Turned user research into product strategy and patent development for a **CES Innovation Award-winning** VR device, analyzing UI/UX feedback from **200+ alpha testers**.

Projects

LLM Financial QA Agent | *Python, FastAPI, SQLite, RDF/SPARQL, BM25, Docker* **Jul 2026 – Sep 2026**

- Raised QA accuracy to **87% (47/54)** with **19/19 correct refusals** by building an LLM agent that federates SQL, a **1.08M-triple** RDF/OWL knowledge graph, and BM25/vector search.
- Identified tool round-trips (**up to +11.2 s** each), not prompt length (flat 2.9–3.8 s from 599 to 13.5K tokens), as the latency bottleneck, and added a **0.8 ms** deterministic retrieval router.

Tessera | *ESP32, C/C++, PN532 NFC, PySide6, AES-256-GCM, SQLCipher* **Mar 2026 – Present**

- Led a team to place **7th of 650+** in the ROK Army Startup Competition by building an offline EHR dog-tag system.
- Engineered ESP32 firmware to read and decrypt compressed AES-256-GCM records in **under 100 ms**, paired with a PySide6/SQLCipher desktop app for offline serial provisioning.

Sisyphus (sisyphusfocus.site) | *C#, .NET 8, WPF, Windows Internals (IFEO)* **Apr 2026 – Present**

- Reached **180+ users** with a focus utility that enforces app and website blocking, logs categorized activity, and surfaces AI-generated feedback and task management.
- Engineered a C#/.NET WPF client with near-zero idle CPU usage, Windows Registry/IFEO-based blocking, an installer, and binary-differencing updates for automated client deployment.

Ajou Course Manager | *Django, Docker, K8s, React, PDFMiner, PostgreSQL* **Oct 2025 – Feb 2026**

- Built a full-stack academic management platform for Ajou Medical School, **handling 70+ courses, 1,400+ lectures and 400+ accounts across 9 user roles,** including Excel import/export, AI-based syllabus extraction and analytics.
- Shipped features requested by faculty and staff, backed by **600+ automated tests**.

Leadership & Activities

2nd Infantry Division Sustainment Brigade **Jun 2025 – Dec 2026**
BDE UMT KATUSA Sergeant (E-5) *Camp Humphreys, South Korea*

- Built internal servers with local AI and autonomous drone systems as a researcher at the 2ID Warrior Works lab.
- Provided tactical translation for the United Nations Command during ROK–U.S. combined exercises, earning the **Army Commendation Medal and Army Achievement Medal**.

Directed Reading Program **Spring 2023 – Spring 2024**
Natural Language Reasoning Team *University of Texas at Austin*

- Led weekly discussions on CoT and LLM reasoning papers, **covering 17 papers with UT CS Ph.D. mentorship**.